

LAB EXPERIMENT RECORD

Set 2: Customer Feedback Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	PRANAYAA S
Register Number	192524110
Date	29.09.2026
Day	Tuesday

Aim

To analyze customer satisfaction data using a histogram, pie chart, and stacked bar chart in R, and to develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

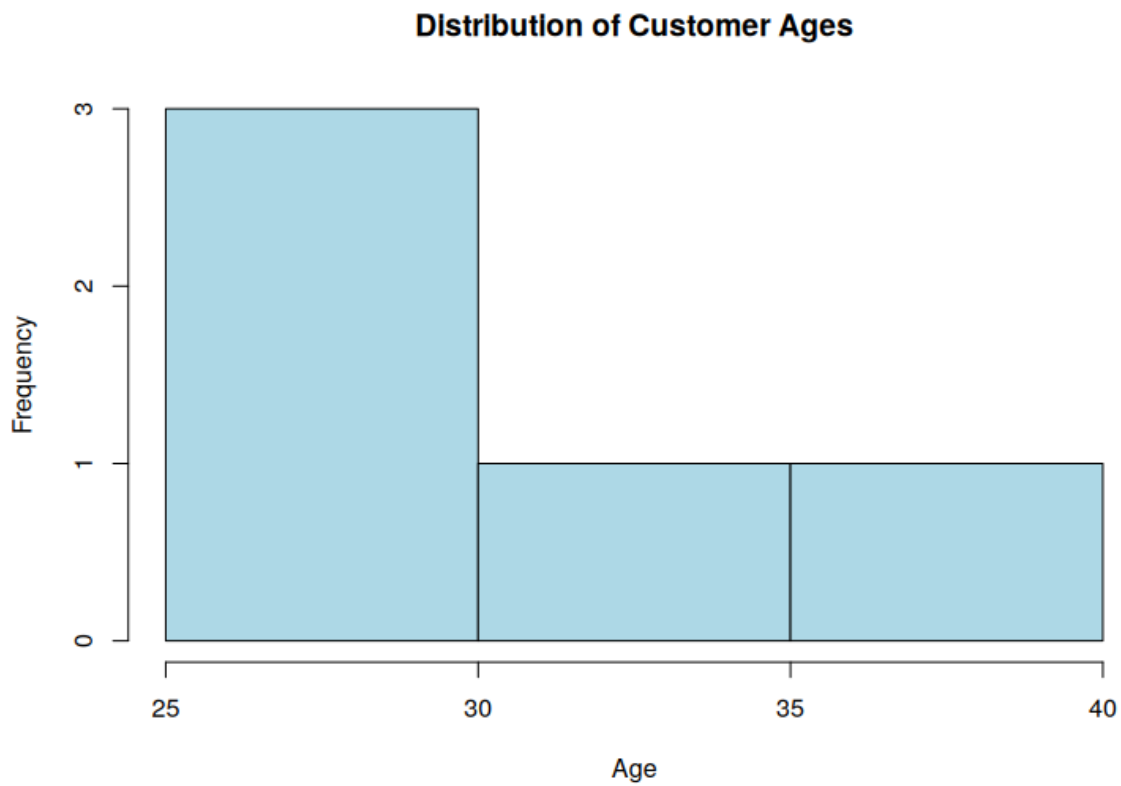
Note: The experiment table does not include feedback text. For Question 4, the R program uses one feedback comment for each customer, as shown in the code.

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Create dataset for Set 2
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5),
  Feedback = c(
    "Great product, excellent quality and fast delivery!",
    "Amazing service, loved the fast delivery and friendly support.",
    "Average experience, product quality could be improved.",
    "Good quality overall, reasonable pricing and good service.",
    "Excellent experience! Highly recommended, super fast delivery."
  )
)
```

```
print(customer_data)
# 1. Histogram for Age Distribution
hist(customer_data$Age,
      col = "lightblue",
      border = "black",
      xlab = "Age",
      ylab = "Frequency",
      main = "Distribution of Customer Ages",
      breaks = 5)
```



Output 1: Distribution of Customer Ages

Observation: Three of the five customers are between 25 and 30 years old, and one customer each falls in the 30–35 and 35–40 ranges. Most customers are in their late twenties.

PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

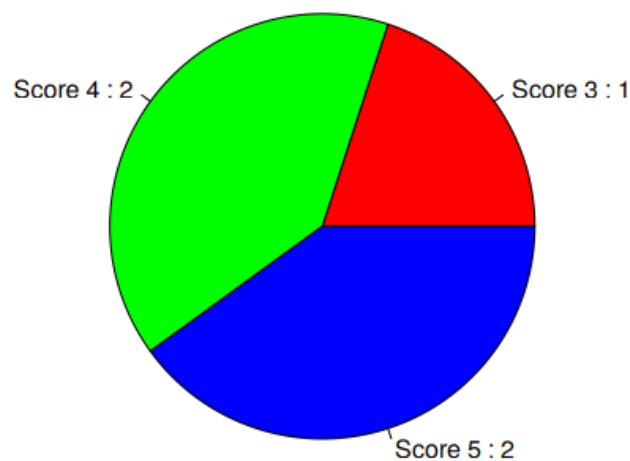
```
# Create dataset for Set 2
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5),
  Feedback = c(
    "Great product, excellent quality and fast delivery!",
    "Amazing service, loved the fast delivery and friendly support.",
    "Average experience, product quality could be improved.",
    "Good quality overall, reasonable pricing and good service.",
    "Excellent experience! Highly recommended, super fast delivery."
  )
)

print(customer_data)

# 2. Pie Chart for Satisfaction Scores
score_counts <- table(customer_data$Satisfaction_Score)
pie_labels <- paste("Score", names(score_counts), ":", score_counts)

pie(score_counts,
    labels = pie_labels,
    col = rainbow(length(score_counts)),
    main = "Distribution of Customer Satisfaction Scores")
```

Distribution of Customer Satisfaction Scores



Output 2: Distribution of Customer Satisfaction Scores

Observation: Scores 4 and 5 each account for two customers (40% each), while only one customer gave a score of 3 (20%). This means 80% of the customers rated their satisfaction as 4 or above.

PROGRAM 3: Stacked Bar Chart of Satisfaction by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

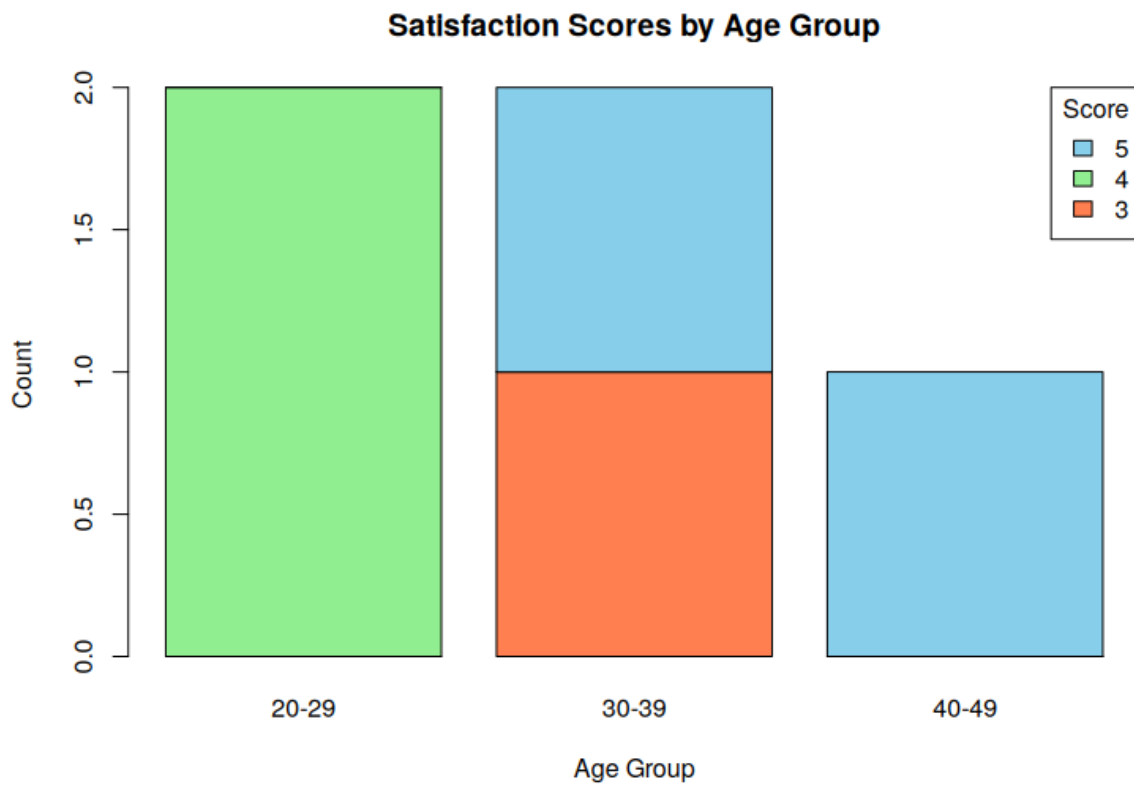
```
# Create dataset for Set 2
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5),
  Feedback = c(
    "Great product, excellent quality and fast delivery!",
    "Amazing service, loved the fast delivery and friendly support.",
    "Average experience, product quality could be improved.",
    "Good quality overall, reasonable pricing and good service.",
    "Excellent experience! Highly recommended, super fast delivery."
  )
)

print(customer_data)

# Categorize ages into groups
customer_data$Age_Group <- cut(customer_data$Age,
                               breaks = c(20, 30, 40, 50),
                               labels = c("20-29", "30-39", "40-49"),
                               right = FALSE)

# Create a contingency table
age_satisfaction_table <- table(customer_data$Satisfaction_Score, customer_data$Age_Group)

# 3. Stacked Bar Chart
barplot(age_satisfaction_table,
        col = c("coral", "lightgreen", "skyblue"),
        xlab = "Age Group",
        ylab = "Count",
        main = "Satisfaction Scores by Age Group",
        legend.text = TRUE,
        args.legend = list(title = "Score", x = "topright"))
```



Output 3: Satisfaction Scores by Age Group

Observation: The 20–29 group has two customers, both with a score of 4. The 30–39 group has one score of 5 and one score of 3, and the 40–49 group has one customer with a score of 5. Every age group has at least one customer scoring 4 or above.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
# install.packages("wordcloud") # run once if wordcloud is not installed
library(wordcloud)

# Create dataset for Set 2
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  Satisfaction_Score = c(4, 5, 3, 4, 5),
  Feedback = c(
    "Great product, excellent quality and fast delivery!",
    "Amazing service, loved the fast delivery and friendly support.",
    "Average experience, product quality could be improved.",
    "Good quality overall, reasonable pricing and good service.",
    "Excellent experience! Highly recommended, super fast delivery."
  )
)

print(customer_data)

# Combine all feedback and clean the text
feedback_text <- tolower(paste(customer_data$Feedback, collapse = " "))
feedback_text <- gsub("[[:punct:]]", "", feedback_text)

# Split into words and remove common stop words
words <- unlist(strsplit(feedback_text, "\\s+"))
stop_words <- c("and", "the", "could", "be", "overall")
words <- words[!(words %in% stop_words) & nchar(words) > 2]

# Word frequency table
word_freq <- sort(table(words), decreasing = TRUE)
print(word_freq)

# 4. Word Cloud of Customer Feedback
set.seed(123)
wordcloud(words = names(word_freq),
  freq = as.numeric(word_freq),
  min.freq = 1,
  scale = c(4, 0.9),
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2"))
title(main = "Customer Feedback Word Cloud")
```

Customer Feedback Word Cloud



Output 4: Customer Feedback Word Cloud

Observation: The most frequent words are "fast", "delivery", and "quality" (3 times each), followed by "excellent", "experience", "good", "product", and "service" (2 times each). Most words are positive, showing that customers value fast delivery and product quality. The word "improved" points to one customer who felt the quality could be better.

Procedure

1. Create the customer data frame with Customer ID, Age, Satisfaction Score, and Feedback.
2. Use `hist()` to show the distribution of customer ages.
3. Use `table()` to count the satisfaction scores and `pie()` to draw the labelled pie chart.
4. Create age groups using `cut()` with `right = FALSE`, form a score × age-group table, and use `barplot()` for the stacked bar chart.
5. Combine and clean the feedback text, split it into words, remove stop words, and count word frequencies with `table()`.
6. Use `wordcloud()` to display the most frequent feedback words.

Result

The customer satisfaction data was successfully visualized using a histogram, pie chart, and stacked bar chart, and a word cloud was created to identify the most common customer sentiments.

Key R Functions Used

Function	Purpose
<code>hist()</code>	Creates the histogram of customer ages.
<code>table()</code>	Counts satisfaction scores and word frequencies.
<code>pie()</code>	Creates the satisfaction score pie chart.
<code>cut()</code>	Divides ages into age groups.
<code>barplot()</code>	Creates the stacked bar chart.
<code>gsub()</code> / <code>strsplit()</code>	Cleans the feedback text and splits it into words.
<code>wordcloud()</code>	Creates the word cloud of feedback.